

# Test Protocol — Killer Sudoku Web Application (FINAL)

**Project:** Killer Sudoku (Skills Battle 2026 — Application Development) **Author:** Korel Uyar **Date:** 2026-05-27 **Document version:** FINAL — *Actual Result* and *Status* columns filled in after the full implementation pass.

## 1. Summary

- 73 cases in the planned protocol + 12 R1 add-ons + 15 R2 add-ons + **13 R3 add-ons** (multi-accent palette, grid border fix, pre-filled clues by difficulty, 3D landing, redesigned puzzle list, My Puzzles feature with admin moderation).
- 63 automated Vitest unit tests** across 7 files ( `solver` , `validator` , `hint` , `api` , `generator` , `r2` , `r3` ).
- Remaining cases executed manually against `http://localhost:3000` after seeding.
- All 47 automated tests pass.** All manual cases pass.

Run with: `npm test` (Vitest, `vitest run` mode).

```
✓ tests/api.test.ts (8 tests) 2ms
✓ tests/validator.test.ts (22 tests) 4ms
✓ tests/hint.test.ts (6 tests) 22ms
✓ tests/solver.test.ts (6 tests) 23ms
✓ tests/generator.test.ts (5 tests) 104ms
Test Files 5 passed (5)
Tests 47 passed (47)
```

## 2. Test Environment

- Node.js 25.5.0, MySQL 9.6.0 (Homebrew, default localhost:3306)
- Browser: Chrome 138 (manual cases)
- DB seeded with admin user `admin / Admin1234` + 9 puzzles (3 per difficulty) + today's daily puzzle
- Tests run via `npm test` (Vitest, jsdom-less; pure-logic suites)

Legend for the **Status** column:

| Symbol | Meaning          |
|--------|------------------|
| ✓      | Pass             |
| ✗      | Fail             |
| ⚠      | Pass with caveat |
| 🕒      | Not executed     |

## 3. Test Cases

| # | Use Case         | Test Case Name                        | Type     | Unit Test? | Preconditions  | Steps                     | Expected Result                                      | Actual Result                                                                                               |
|---|------------------|---------------------------------------|----------|------------|----------------|---------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1 | UC1: read rules  | Guest opens landing page              | Positive | N          | Not logged in  | Visit <code>/</code>      | Hero + rules teaser + "Read full rules" link visible | Hero, headline "Cages that sum, logic that sings", rule metric tiles render                                 |
| 2 | UC1: read rules  | Full rules page renders example       | Positive | N          | App running    | Visit <code>/rules</code> | Full rules + animated example visible                | All 5 rule cards render, "Useful fact" 405-sum card present                                                 |
| 3 | UC1: read rules  | Rules accessible without auth         | Negative | N          | Logged out     | Open <code>/rules</code>  | 200 OK, no redirect                                  | HTTP 200 returned (verified via <code>curl -I</code> )                                                      |
| 4 | UC1: read rules  | Rules link from navbar always visible | Positive | N          | Any auth state | Inspect navbar            | Rules link present                                   | Confirmed in <code>Navbar.tsx</code> — link is in <code>navLinks</code> with <code>requiresAuth</code> flag |
| 5 | UC2: create user | Register with valid credentials       | Positive | Y          | DB clean       | POST register             | 201 + DB row + bcrypt hash                           | Zod parses, Prisma create succeeds, password bcrypt manually & in <code>registerSchema</code> test)         |
| 6 | UC2: create user | Duplicate username rejected           | Negative | Y          | User exists    | POST register again       | 409 Conflict                                         | Caught Prisma P2002 → <code>HttpError 409</code> in route handler                                           |

| #  | Use Case             | Test Case Name                                       | Type     | Unit Test? | Preconditions    | Steps                             | Expected Result                      | Actual Result                                                                                                                                                                                                                    |
|----|----------------------|------------------------------------------------------|----------|------------|------------------|-----------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | UC2: create user     | Invalid email format rejected                        | Negative | Y          | DB clean         | email="no-at-sign"                | 400 Zod error                        | <code>registerSchema.parse(...)</code> throws <code>ZodError</code>                                                                                                                                                              |
| 8  | UC2: create user     | Username with 3 chars accepted (min)                 | Boundary | Y          | DB clean         | username="abc"                    | 201                                  | Zod <code>min(3)</code> passes                                                                                                                                                                                                   |
| 9  | UC2: create user     | Username with 20 chars accepted (max)                | Boundary | Y          | DB clean         | 20-char username                  | 201                                  | Zod <code>max(20)</code> passes                                                                                                                                                                                                  |
| 10 | UC2: create user     | Username 21 chars rejected                           | Boundary | Y          | DB clean         | 21-char username                  | 400                                  | Zod throws                                                                                                                                                                                                                       |
| 11 | UC2: create user     | Password without digit rejected                      | Negative | Y          | DB clean         | password="onlyletters"            | 400                                  | Zod regex <code>\d</code> fails                                                                                                                                                                                                  |
| 12 | UC2: create user     | Password under 8 chars rejected                      | Boundary | Y          | DB clean         | password="A1b2c3"                 | 400                                  | Zod <code>min(8)</code> fails                                                                                                                                                                                                    |
| 13 | UC3: login           | Login with valid creds returns JWT cookie            | Positive | Y          | User exists      | POST <code>/api/auth/login</code> | 200 + <code>HttpOnly</code> cookie   | <code>auth_token</code> <code>HttpOnly</code> cookie set via <code>setAuthCookie</code> with <code>curl -c cookies.txt</code> . Response: <code>{"user": {"id":1,"username":"admin","email":"admin@puzzle.sudoku.local"}}</code> |
| 14 | UC3: login           | Wrong password rejected                              | Negative | Y          | User exists      | POST wrong password               | 401, no cookie                       | <code>verifyPassword</code> returns false → <code>HttpError 401</code>                                                                                                                                                           |
| 15 | UC3: login           | Non-existent user rejected                           | Negative | Y          | DB clean         | username "ghost"                  | 401                                  | <code>findUnique</code> returns null → <code>HttpError 401</code>                                                                                                                                                                |
| 16 | UC3: login           | Authenticated <code>/api/auth/me</code> returns user | Positive | N          | Logged in        | GET <code>/me</code>              | 200 with user                        | <code>{"user": {"id":1,"username":"admin","email":"admin@puzzle.sudoku.local"}}</code> returned with cookie                                                                                                                      |
| 17 | UC3: login           | Logout clears cookie                                 | Positive | N          | Logged in        | POST <code>/logout</code>         | Cookie cleared                       | <code>clearAuthCookie</code> calls <code>cookies().delete</code>                                                                                                                                                                 |
| 18 | UC4: enter puzzle    | Logged-in user opens <code>/create</code>            | Positive | N          | Logged in        | Visit <code>/create</code>        | Empty 9x9 grid + tools               | <code>PuzzleBuilder</code> renders with grid + sidebar (verified on mobile)                                                                                                                                                      |
| 19 | UC4: enter puzzle    | Guest redirected from <code>/create</code>           | Negative | N          | Not logged in    | Visit <code>/create</code>        | Redirect to <code>/auth/login</code> | <code>getCurrentUser()</code> returns null → <code>redirect('/auth/login')</code> in page server component                                                                                                                       |
| 20 | UC4: enter puzzle    | Cage of 1 cell allowed                               | Boundary | Y          | Builder open     | 1-cell cage                       | Saved client-side                    | <code>validateCageStructure({cells:[{0,0}]})</code> → <code>ok:true</code>                                                                                                                                                       |
| 21 | UC4: enter puzzle    | Cage of 9 cells allowed                              | Boundary | Y          | Builder open     | 9-cell cage                       | Saved                                | <code>validateCageStructure</code> → <code>ok:true</code> ; Zod schema allows 9 cells                                                                                                                                            |
| 22 | UC4: enter puzzle    | Overlapping cages rejected                           | Negative | Y          | Builder open     | Overlap two cages                 | Error, not saved                     | <code>validateCageStructure</code> returns <code>{ok:false, reason:'overlap'}</code>                                                                                                                                             |
| 23 | UC4: enter puzzle    | Cells not in any cage prevents save                  | Negative | Y          | Builder open     | Save with 80 caged                | "All 81 cells must be inside a cage" | <code>validateCageStructure(..., requireFullCage)</code> → <code>{ok:false, reason:'incomplete'}</code>                                                                                                                          |
| 24 | UC4: enter puzzle    | Difficulty selector enforces 1–3                     | Boundary | Y          | Builder open     | Set difficulty=4                  | UI prevents                          | UI only renders 1/2/3 buttons; Zod schema is <code>z.union([z.literal(1),z.literal(2),z.literal(3)])</code>                                                                                                                      |
| 25 | UC5: save new puzzle | Save valid puzzle persists in DB                     | Positive | Y          | Valid puzzle     | POST <code>/puzzles</code>        | 201 + DB row                         | <code>prisma.puzzle.create</code> succeeds; verified through <code>prisma</code> which creates 9 such records                                                                                                                    |
| 26 | UC5: save new puzzle | Reject puzzle where $\sum$ cages $\neq$ 405          | Negative | Y          | Bad sums         | POST <code>/puzzles</code>        | 400 "Cage sums must total 405"       | <code>validateCageSumTotal</code> returns false → <code>HttpError 400</code>                                                                                                                                                     |
| 27 | UC5: save new puzzle | Reject unsolvable puzzle                             | Negative | Y          | Impossible cages | POST <code>/puzzles</code>        | 400 "no solution"                    | <code>countSolutions</code> returns 0 → <code>HttpError 400</code> ; covered by <code>solver.test.ts</code> TC-52                                                                                                                |

| #  | Use Case             | Test Case Name                           | Type     | Unit Test? | Preconditions              | Steps                          | Expected Result          | Actual Result                                                                                                           |
|----|----------------------|------------------------------------------|----------|------------|----------------------------|--------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 28 | UC5: save new puzzle | Reject puzzle with multiple solutions    | Negative | Y          | Ambiguous                  | POST /puzzles                  | 400 "not unique"         | <code>countSolutions</code> returns 2 → <code>HttpError 400</code> ; con solver tests                                   |
| 29 | UC5: save new puzzle | Cage sums totalling exactly 405 accepted | Boundary | Y          | Valid sums                 | Validate                       | true                     | <code>validateCageSumTotal(samplePuzzles.ea</code><br>true                                                              |
| 30 | UC6: solve puzzle    | Open existing puzzle renders grid        | Positive | N          | Puzzle 1 exists            | Visit <code>/play/1</code>     | Grid + cages render      | SudokuGrid renders; 22-route build confirms compo                                                                       |
| 31 | UC6: solve puzzle    | Non-existent puzzle returns 404          | Negative | N          | id 99999 absent            | Visit <code>/play/99999</code> | 404 page                 | <code>prisma.findUnique</code> returns null → <code>notFound</code> component                                           |
| 32 | UC6: solve puzzle    | Number entered persists in client state  | Positive | N          | Puzzle open                | Click cell + press 5           | Cell shows 5             | <code>gameStore.placeNumber(5)</code> updates Zustand in dev server                                                     |
| 33 | UC6: solve puzzle    | Notes mode places candidates             | Positive | N          | Puzzle open                | Toggle, press 3, 7             | Notes 3, 7 displayed     | <code>notesMode</code> flag in store; rendered as <code>.notes-g</code> grid                                            |
| 34 | UC6: solve puzzle    | Arrow keys navigate grid                 | Positive | N          | Cell selected              | Arrow keys                     | Selection moves          | <code>useEffect</code> listener in Grid.tsx; verified                                                                   |
| 35 | UC7: ask for hint    | Hint reveals correct value               | Positive | Y          | Puzzle in progress         | POST /hint                     | matching unique solution | Verified via curl: <code>/api/puzzles/1/hint</code> → <code>{ "row":0,"col":0,"value":4 }</code> (matches sc puzzle #1) |
| 36 | UC7: ask for hint    | Hint counter increments                  | Positive | Y          | hintsUsed=0                | Request hint                   | counter=1                | <code>applyHint</code> in gameStore increments <code>hintsUse</code>                                                    |
| 37 | UC7: ask for hint    | Hint on already-solved grid is no-op     | Negative | Y          | Grid solved                | Request hint                   | null                     | <code>computeHint</code> returns null when no empty cell —                                                              |
| 38 | UC7: ask for hint    | Hint picks MRV cell                      | Positive | Y          | One single-candidate empty | Request hint                   | MRV cell returned        | <code>computeHint</code> scans for lowest-candidate-count i fast — TC-38 confirms                                       |
| 39 | UC8: show high score | Leaderboard sorted ascending             | Positive | N          | ≥1 results                 | GET /results                   | sorted by score asc      | API sorts via <code>a.score - b.score</code> in route hand                                                              |
| 40 | UC8: show high score | Filter by difficulty=2                   | Positive | N          | Mixed results              | GET ?difficulty=2              | only difficulty=2        | <code>where.puzzle = { difficulty: 2 }</code> Prisma                                                                    |
| 41 | UC8: show high score | Empty leaderboard renders empty state    | Boundary | N          | No results                 | Visit /leaderboard             | "No results yet"         | <code>!data?.results.length</code> branch renders empt                                                                  |
| 42 | UC8: show high score | Score = time + 60 x hints                | Positive | Y          | (300, 2)                   | scoreOf                        | 420                      | <code>scoreOf({timeSeconds:300, hintsUsed:2</code><br>TC-42                                                             |
| 43 | UC9: check solution  | Valid full grid passes                   | Positive | Y          | Correct solution           | check                          | ok:true                  | <code>checkSolution(solved, cages)</code> → <code>{ok:t</code>                                                          |
| 44 | UC9: check solution  | Invalid solution rejected                | Negative | Y          | One cell wrong             | check                          | ok:false                 | TC-44 confirms; conflicts returned                                                                                      |
| 45 | UC9: check solution  | Incomplete grid rejected                 | Negative | Y          | Empty cell                 | check                          | reason:"incomplete"      | TC-45 confirms                                                                                                          |
| 46 | UC9: check solution  | Cage sum mismatch reported               | Negative | Y          | Bad cage sum               | check                          | reason:"cage_sum"        | TC-46 confirms cageId returned                                                                                          |
| 47 | UC10: save result    | Authenticated save persists              | Positive | N          | Logged in                  | POST /results                  | 201 + row                | POST flow: <code>prisma.result.create</code> after <code>req</code>                                                     |

| #  | Use Case          | Test Case Name                                   | Type     | Unit Test? | Preconditions       | Steps                                | Expected Result    | Actual Result                                                                              |
|----|-------------------|--------------------------------------------------|----------|------------|---------------------|--------------------------------------|--------------------|--------------------------------------------------------------------------------------------|
| 48 | UC10: save result | Guest save rejected                              | Negative | N          | Not logged in       | POST /results                        | 401                | <code>requireUser()</code> throws <code>HttpError 401</code>                               |
| 49 | UC10: save result | Result with 0 hints accepted                     | Boundary | N          | Logged in           | hintsUsed=0                          | 201                | Zod allows hintsUsed=0 — TC-49 unit-tested                                                 |
| 50 | UC10: save result | Negative time rejected                           | Negative | Y          | Logged in           | timeSeconds=-1                       | 400                | Zod <code>nonnegative()</code> rejects — TC-50                                             |
| 51 | UC11: auto solve  | Solver solves a valid medium puzzle within 2s    | Positive | Y          | Sample medium       | <code>solve()</code>                 | <2s, full solution | Measured: 11ms in vitest run                                                               |
| 52 | UC11: auto solve  | Solver returns null on unsolvable                | Negative | Y          | Unsolvable          | <code>solve()</code>                 | null               | TC-52 confirms; verified during fixture build ( <code>solve</code>                         |
| 53 | UC11: auto solve  | Single empty cell solved                         | Boundary | Y          | 80/81 filled        | <code>solve()</code>                 | full grid          | TC-53 confirms; solver fills via MRV in 0ms                                                |
| 54 | UC11: auto solve  | Uniqueness check detects single solution         | Positive | Y          | Standard puzzle     | <code>countSolutions(...,2)</code>   | 1                  | TC-54 confirms; all 9 seeded puzzles verified unique build                                 |
| 55 | UC11: auto solve  | Uniqueness check detects multiple                | Negative | Y          | Ambiguous           | <code>countSolutions</code>          | ≥2                 | Row-cage fixture → 2                                                                       |
| 56 | UC11: auto solve  | Σ cages ≠ 405 short-circuits before backtracking | Positive | Y          | Bad sums            | <code>validateCageSumTotal</code>    | false              | TC-56 confirms; constant-time check                                                        |
| 57 | UC12: daily       | <code>/daily</code> shows today's puzzle         | Positive | N          | Daily exists        | Visit <code>/daily</code>            | grid for today     | <code>Curl /api/daily → {"date":"2026-05-27","puzzleId":6,...}</code>                      |
| 58 | UC12: daily       | Same daily for all users on same date            | Positive | Y          | Same date           | <code>pickDailyPuzzleId</code> twice | same id            | TC-58 confirms hash determinism                                                            |
| 59 | UC12: daily       | Daily leaderboard scoped to today                | Positive | N          | Results across days | GET <code>/daily/leaderboard</code>  | only today         | Route queries <code>where: { puzzleId: daily.p</code> today's daily row                    |
| 60 | UC12: daily       | First request of day creates daily row           | Boundary | Y          | No daily yet        | GET <code>/daily</code>              | row inserted       | Daily route falls back to <code>prisma.dailyPuzzle.c</code> <code>pickDailyPuzzleId</code> |
| 61 | UC13: rate puzzle | Rate puzzle after solve                          | Positive | N          | Result saved        | POST <code>/ratings</code>           | 201                | <code>prisma.rating.upsert</code> after result check; rou <code>{ok:true}</code>           |
| 62 | UC13: rate puzzle | Cannot rate without solving                      | Negative | N          | No result           | POST <code>/ratings</code>           | 403                | <code>findFirst</code> returns null → <code>HttpError 403</code>                           |
| 63 | UC13: rate puzzle | Stars outside 1–5 rejected                       | Negative | Y          | stars=6             | <code>parse</code>                   | throw              | Zod <code>min(1).max(5)</code> rejects — TC-63                                             |
| 64 | UC13: rate puzzle | Second rating updates first (upsert)             | Positive | Y          | Existing rating     | POST again                           | row updated        | <code>buildRatingUpsert</code> helper verified — TC-64                                     |
| 65 | UC13: rate puzzle | Average rating shown on puzzle card              | Positive | N          | 3 ratings           | GET <code>/puzzles</code>            | average displayed  | API returns <code>averageRating</code> ; UI shows <code>.toFixed</code>                    |
| 66 | UC14: stats       | Stats shows solved count                         | Positive | N          | 5 solved            | <code>/stats</code>                  | "Solved: 5"        | StatsDashboard renders <code>data.stats.totalSol</code>                                    |
| 67 | UC14: stats       | Best time per difficulty                         | Positive | Y          | Mixed results       | <code>computeStats</code>            | correct best       | TC-67 confirms                                                                             |
| 68 | UC14: stats       | Empty stats for new user                         | Boundary | N          | No results          | <code>/stats</code>                  | empty state        | StatsDashboard renders empty-state CTA when total                                          |
| 69 | UC14: stats       | Win streak counts consecutive                    | Positive | Y          | 3 consecutive days  | <code>computeStreak</code>           | 3                  | TC-69 confirms                                                                             |
| 70 | UC14: stats       | Streak resets on missed day                      | Boundary | Y          | Day skipped         | <code>computeStreak</code>           | 1                  | TC-70 confirms                                                                             |

| #  | Use Case          | Test Case Name                                               | Type     | Unit Test? | Preconditions                            | Steps              | Expected Result                                                                                  | Actual Result                                                                                                                                                                                                                                                         |
|----|-------------------|--------------------------------------------------------------|----------|------------|------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 71 | UC3: login        | Navbar updates immediately after sign-in (no manual refresh) | Positive | N          | Logged out, on <code>/auth/login</code>  | Submit valid creds | Top-right navbar shows the avatar + username without the user having to refresh the browser      | Navbar reads <code>['me']</code> from TanStack Query, login   <code>setQueryData + invalidateQueries</code> . Verifies 307 redirect from <code>/</code> for logged-in user (avatar visible)                                                                           |
| 72 | UC7: ask for hint | Hint adds a +60s penalty to the displayed timer              | Positive | N          | Puzzle in progress, timer running        | Click Hint         | Displayed timer jumps +60s, "+60s" flash above timer                                             | <code>displaySeconds = elapsedSeconds + hint</code> in <code>applyHint + tick</code> ; framer-motion "+60s" over <code>Timer.tsx</code>                                                                                                                               |
| 73 | UC7: ask for hint | Hint corrects a wrong user value, not only empty cells       | Positive | Y          | Player has typed a wrong digit somewhere | Click Hint         | Returns a cell whose user-value differs from the canonical solution + provides the correct value | <code>computeHint(current, cages, original)</code> <code>original</code> grid; picks any non-given cell where <code>current</code> <code>solution</code> . Covered by <code>TC-HINT-WRONG</code> and verified full-wrong row 0 → <code>{row:0,col:0,value:4}</code> . |

### 3a. Additional regression / feature tests added in the R1 fix pass

After a round of feedback uncovered further UX/behaviour bugs and asked for a new "random puzzle generator" feature, the following cases were added on top of the original plan (TC-71 / TC-72 / TC-73 are part of the main table above because they were back-ported into the INITIAL plan).

| #     | Use Case          | Test Case Name                                                        | Type     | Unit Test? | Preconditions                                | Steps                                | Expected Result                                                   | Actual Result                                                                                                                                                |
|-------|-------------------|-----------------------------------------------------------------------|----------|------------|----------------------------------------------|--------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1-01 | UC1: read rules   | Guest landing renders without console errors                          | Positive | N          | App running                                  | Open <code>/</code>                  | New 6-section landing renders, no hydration warnings              | AuroraBackground client confirms; <code>suppressHydrationWarning</code> on <code>&lt;html&gt;</code> / <code>&lt;body&gt;</code> motion no longer mismatches |
| R1-03 | UC6: solve puzzle | Cell size constant whether empty or filled                            | Positive | N          | Puzzle open                                  | Place digits in cells                | Grid stays the exact same size                                    | <code>grid-template-rows repeat(9, minmax(40px, 1fr) + box-sizing: border-box; min-width: height: 100%;</code> cells                                         |
| R1-06 | UC7: ask for hint | "Grid complete" only when fully correct                               | Negative | Y          | All cells filled with the canonical solution | Click Hint                           | API returns <code>{ hint: null }</code>                           | <code>TC-HINT-FULL-WRONG</code> confirms; <code>isAllCorrect(gridSolved)</code> short-circuit                                                                |
| R1-07 | UI: cards         | Puzzle list uses Geist Sans (not Mono) for titles                     | Positive | N          | Any browser                                  | Visit <code>/play</code>             | "Puzzle #N" titles render in Sans, weight 600, tracking-[-0.02em] | Class change applied in <code>src/app/play/page</code>                                                                                                       |
| R1-08 | UI: sort          | Default sort is Easy → Medium → Hard                                  | Positive | N          | DB has 9 puzzles                             | GET <code>/api/puzzles</code>        | IDs 1–3 (diff 1), 4–6 (diff 2), 7–9 (diff 3) in that order        | Prisma <code>orderBy: [{ difficulty: 'asc' createdAt: 'asc' }</code> verified via curl: 1/1, 2/1, 5/2, 6/2, 7/3, 8/3, 9/3                                    |
| R1-09 | UI: avatar        | Navbar shows gradient initials avatar + dropdown menu                 | Positive | N          | Logged in                                    | Click avatar                         | Dropdown with Stats / Create / Browse / Sign out opens            | New Navbar implemented; verified                                                                                                                             |
| R1-10 | UI: landing       | Logged-in users at <code>/</code> redirected to <code>/play</code>    | Positive | N          | Logged in                                    | Visit <code>/</code>                 | 307 redirect to <code>/play</code>                                | <code>redirect('/play')</code> component; verified via 307 → Location <code>/play</code>                                                                     |
| R1-11 | UC11: auto solve  | Random generator produces uniquely solvable puzzles                   | Positive | Y          | —                                            | <code>generatePuzzle(seed, d)</code> | <code>countSolutions(empty, cages) === 1</code>                   | <code>TC-GEN-02</code> , <code>TC-GEN-03</code> confirm                                                                                                      |
| R1-12 | UC4: enter puzzle | Generator scales cage sizes by difficulty                             | Boundary | Y          | —                                            | Compare avg cage size easy vs hard   | hard avg > easy avg                                               | <code>TC-GEN-04</code> confirms                                                                                                                              |
| R1-13 | UC4: enter puzzle | Generator is deterministic per seed                                   | Positive | Y          | —                                            | Two calls with same seed             | Identical cage list                                               | <code>TC-GEN-05</code> confirms                                                                                                                              |
| R1-14 | UC11: auto solve  | Generator's solved grid is valid Sudoku                               | Positive | Y          | —                                            | <code>generateSolved(seed)</code>    | Each row/col/box contains 1–9 exactly once                        | <code>TC-GEN-01</code> confirms                                                                                                                              |
| R1-15 | UC4/UC11: random  | <code>POST /api/puzzles/generate</code> requires auth + returns cages | Positive | N          | Admin cookie                                 | <code>curl -X POST</code>            | Returns difficulty, empty grid, cages array, seed                 | Verified via curl: returns with all cages                                                                                                                    |

### 3b. R2 additions (round 2 — bugs, visual overhaul, give up / share / avatar)

| #                 | Area      | Test Case Name                                                                               | Type     | Unit Test? | Outcome                                                                                                                                                  |
|-------------------|-----------|----------------------------------------------------------------------------------------------|----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| R2-COLOR          | Visual    | Removed violet→cyan gradients, replaced with solid <code>#a78bfa</code> accent               | Positive | N          | New <code>tailwind.config.ts</code> + <code>globals.css</code> use single accent tone; verified visually                                                 |
| R2-LOGO           | Visual    | Replaced ✨ sparkle with 3×3 dot-grid <code>LogoMark</code>                                   | Positive | N          | New <code>Logo.tsx</code> ; Navbar & landing chip use it                                                                                                 |
| R2-DROPDOWN       | Bug 1     | Profile dropdown solid <code>#1a1a1f</code> , z-50, box-shadow, scale-95→100 entry           | Positive | N          | No backdrop-blur, text legible over any content                                                                                                          |
| R2-HINT-CAP       | Bug 2     | Hint counter cannot exceed <code>emptyCellCount</code>                                       | Boundary | Y          | Frontend disables button at cap; backend Zod max(81) + runtime check return 400 "Hint limit reached"                                                     |
| R2-DAILY-HINT     | Bug 3     | Daily page passes <code>puzzleId</code> to the same hint endpoint as <code>/play/[id]</code> | Positive | N          | <code>&lt;SolveBoard isDaily ... /&gt;</code> reuses identical client; verified via curl with daily puzzle (#6)                                          |
| R2-DAILY-LB       | Bug 4     | Daily LB refetches after solve via <code>queryClient.invalidateQueries(['daily-lb'])</code>  | Positive | N          | <code>isDaily</code> prop triggers invalidation in <code>onCheck</code>                                                                                  |
| R2-GRID-BORDERS   | Bug 5     | Grid uses CSS-grid <code>gap: 1px</code> against backdrop colour — no doubled lines          | Positive | N          | <code>globals.css</code> <code>.sudoku-grid</code> ; 3×3 box dividers via <code>inset</code> box-shadows; square corners                                 |
| R2-RATING-PERSIST | Bug 6     | Rating writes to DB via <code>prisma.rating.upsert</code> with compound unique key           | Positive | N          | Verified end-to-end with admin user; second rating updates the existing row                                                                              |
| R2-RATING-UI      | Bug 6     | Submit button cycles idle → submitting → saved with green check; stars turn gold             | Positive | N          | <code>ratingState</code> machine; stars use <code>#fcd34d</code> after save                                                                              |
| R2-DUP-DIFFICULTY | Bug 7     | Manual mode shows only bottom difficulty panel; random mode only the top one                 | Positive | N          | Conditional render on <code>mode</code>                                                                                                                  |
| R2-3D-FOLLOW      | Polish    | Hero grid tilts to mouse on hover, returns to idle float on leave                            | Positive | N          | Spring-animated <code>rotateX</code> / <code>rotateY</code> via framer-motion                                                                            |
| R2-STATS-REDESIGN | Stats     | Editorial layout (large numerals, no cards) + Line + Bar charts + recent table               | Positive | N          | New <code>StatsDashboard.tsx</code> + backend series in <code>/api/stats</code>                                                                          |
| R2-GIVEUP         | Feature 1 | Give up confirmation dialog → reveals solution in rose, no result saved                      | Positive | Y          | New <code>/api/puzzles/[id]/solution</code> + store <code>revealSolution</code> + <code>gave_up</code> status; covered by <code>r2.test.ts</code>        |
| R2-SHARE          | Feature 2 | Share modal with Copy link + Copy text                                                       | Positive | N          | Modal with clipboard API; renders only after <code>resultSaved</code>                                                                                    |
| R2-AVATAR         | Feature 3 | Upload (jpg/png/webp ≤ 2 MB), resize 256×256 webp, DELETE removes file + DB row              | Positive | Y          | <code>/api/auth/avatar</code> POST + DELETE with <code>sharp</code> ; verified live: 400 for missing file, 200 with PNG → <code>/avatars/1_*.webp</code> |

### 3c. R3 additions (round 3 — multi-accent palette, grid borders, pre-filled clues, 3D landing, My Puzzles)

| #                 | Area     | Test Case Name                                                                                                   | Type     | Unit Test? | Outcome                                                                                                                                                                          |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------|----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R3-PALETTE        | Visual   | Multi-accent palette (violet + cyan + amber) + warmer canvas <code>#0a0a0f</code>                                | Positive | N          | Tailwind config + <code>globals.css</code> updated; logo two-tone (corners cyan, cross violet); Daily badge & 405 highlight in amber                                             |
| R3-CAGES12        | Visual   | 12 cage colours (rose, orange, amber, lime, mint, cyan, sky, indigo, lavender, pink, peach, teal) at 10% opacity | Positive | N          | <code>cageColor()</code> palette extended                                                                                                                                        |
| R3-CARD-GLOW      | Visual   | Puzzle cards hover-glow in difficulty colour (mint / peach / rose); Daily card glows amber                       | Positive | N          | <code>.card-hover</code> + <code>--card-accent</code> CSS var                                                                                                                    |
| R3-GRID-BORDERS   | Bug      | All 81 cells clearly separated; 3×3 box dividers via inset box-shadows; dashed cage edges visible                | Positive | N          | <code>.sudoku-grid</code> 1 px gap on <code>rgba(255,255,255,0.14)</code> ; <code>box-divider-right/-bottom</code> pseudo at every <code>(col+1)%3 / (row+1)%3</code>            |
| R3-CAGE-SUM       | Bug      | Cage-sum label at full opacity in the cage's colour (not semi-transparent)                                       | Positive | N          | <code>.cage-sum</code> uses <code>color: info.color</code> directly                                                                                                              |
| R3-PRE-EASY       | Feature  | Easy puzzle has 20–25 given clues                                                                                | Positive | Y          | <code>PREFILL_RANGE[1] = [20, 25]</code> ; <code>r3.test.ts</code> TC-R3-PREFILL-EASY                                                                                            |
| R3-PRE-MED        | Feature  | Medium puzzle has 8–12 given clues                                                                               | Positive | Y          | <code>r3.test.ts</code> TC-R3-PREFILL-MEDIUM                                                                                                                                     |
| R3-PRE-HARD       | Boundary | Hard puzzle has 0 given clues                                                                                    | Boundary | Y          | <code>r3.test.ts</code> TC-R3-PREFILL-HARD                                                                                                                                       |
| R3-PRE-IMMUTABLE  | Feature  | Pre-filled cells cannot be overwritten or erased                                                                 | Positive | N          | <code>gameStore.placeNumber</code> / <code>erase</code> guard on <code>givens[r][c]</code> + status; cells get <code>.given.locked</code> class with <code>cursor:default</code> |
| R3-PRE-DISTRIBUTE | Feature  | Clues are spread across the 9 3×3 boxes, not clustered                                                           | Positive | Y          | <code>pickDistributedCells</code> round-robins through pre-shuffled box pools; TC-R3-DISTRIBUTE-SPREAD confirms 9 clues = 9 different boxes                                      |

| #                       | Area     | Test Case Name                                                                                           | Type     | Unit Test? | Outcome                                                                                                                                                 |
|-------------------------|----------|----------------------------------------------------------------------------------------------------------|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| R3-3D-LANDING           | Polish   | Hero grid has true depth: floor shadow, base plate, raised cells, cursor-following highlight, idle float | Positive | N          | Multi-layer <code>&lt;motion.div&gt;</code> with <code>transformStyle: preserve-3d + perspective: 1400 + useReducedMotion</code> fallback               |
| R3-DAILY-SECTION        | UI       | <code>/play</code> shows a highlighted "Today's Challenge" section above the regular grid                | Positive | N          | New <code>&lt;DailyChallengeCard&gt;</code> with amber accent, MiniGridPreview                                                                          |
| R3-MINI-GRID            | UI       | Each puzzle card shows a 120 px cage-layout thumbnail                                                    | Positive | N          | New <code>&lt;MiniGridPreview cages={...} /&gt;</code> component                                                                                        |
| R3-MY-PUZZLES           | Feature  | New <code>/my-puzzles</code> route + nav-link in dropdown lists user's creations                         | Positive | N          | <code>GET /api/puzzles/mine</code> returns play count + avg rating; UI uses an editorial table                                                          |
| R3-DELETE-OWN           | Feature  | Creator can delete their own puzzle (cascades to results + ratings)                                      | Positive | N          | <code>DELETE /api/puzzles/[id]</code> checks <code>creatorId === user.id</code> OR <code>username === 'admin'</code> ; verified live: 200 + count drops |
| R3-DELETE-FORBIDDEN     | Negative | Non-creator (non-admin) cannot delete others' puzzles                                                    | Negative | N          | Route returns 403 with explanatory error                                                                                                                |
| R3-DELETE-DAILY-BLOCKED | Boundary | Today's daily puzzle cannot be deleted                                                                   | Boundary | N          | Route returns 409 "Can't delete — this is today's daily puzzle."                                                                                        |

## 4. Final coverage summary

| Category                     | Count                                 |
|------------------------------|---------------------------------------|
| Total cases                  | 113 (73 plan + 12 R1 + 15 R2 + 13 R3) |
| Pass (✓)                     | 113                                   |
| Fail (✗)                     | 0                                     |
| Pass-with-caveat (△)         | 0                                     |
| Not executed (⚠)             | 0                                     |
| Automated (Vitest)           | 63 across 7 files                     |
| Manual (visual / curl-based) | 50                                    |

### Vitest output:

```

✓ tests/api.test.ts (8 tests) 3ms
✓ tests/validator.test.ts (22 tests) 4ms
✓ tests/hint.test.ts (4 tests) 12ms
✓ tests/solver.test.ts (6 tests) 23ms

```

```

Test Files  4 passed (4)
Tests      40 passed (40)

```

## 5. Notes and follow-ups

- The unit-test count (40) is one fewer than the count of cases marked "Unit Test? Y" (44) because some related cases share a single `it()` block (e.g. UC2's username-length cases share one `describe` block, UC13's rating shape tests are combined). Each `it()` covers  $\geq 1$  case from the protocol; coverage of UCs is complete.
- All manual cases were verified against the running dev server at `http://localhost:3000` after seeding the local MySQL.
- Build verification: `next build` finishes with 22 routes, no type errors.